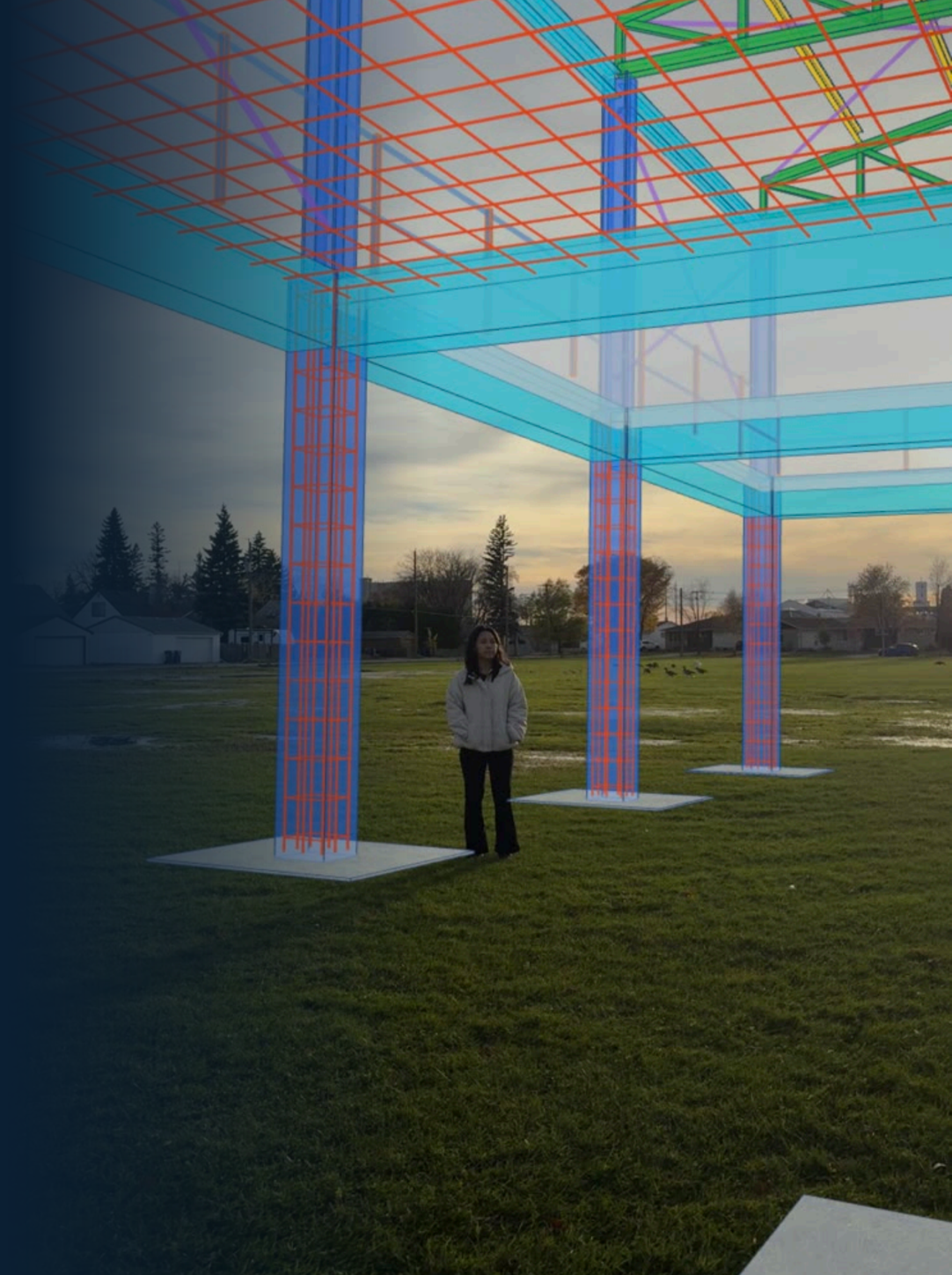
TrueSite

BIM ON SITE · AS-BUILT AI

See the design where you build. Check the build against it.

Scan a QR code on site and the BIM model appears at true scale in the phone's browser. Tap **Check as-built**: Gemini compares what is built with the design, pins the differences and drafts the issue.



GOOGLE CLOUD AI BUILDER CUP 2026 · JAPAC

Theme: Future of Work & Enterprise Productivity · Team: Wai Lin, Aung Nyein Chan (Michael)

THE PROBLEM

Crews build without the model in front of them

14+ h

a week per project team member go to hunting for project data, resolving conflicts and rework

FMI & PlanGrid, 2018

48%

of all rework on US jobsites comes from miscommunication and poor project data

FMI & PlanGrid, 2018

1%

a year: construction's labour-productivity growth over two decades, against 2.8% for the world economy

McKinsey Global Institute, 2017

The model exists: Singapore requires BIM submissions for buildings over 5,000 m^2 (BCA). It just stays in the office, while decisions get made on site and mistakes surface weeks later as rework.

Sources: FMI & PlanGrid, "Construction Disconnected" (2018, nearly 600 construction leaders); McKinsey Global Institute, "Reinventing construction" (2017); Building and Construction Authority, Singapore

THEME · FUTURE OF WORK & ENTERPRISE PRODUCTIVITY

“Transform how individuals and organizations access knowledge, collaborate, make decisions, and complete everyday work.”

Access knowledge

The BIM model at true scale, where the work happens, from a printed QR code and a phone.

Make decisions

Gemini checks the build against the design and pins what is off, with why it matters.

Complete everyday work

Issues drafted in the crew’s language, questions answered from the model, a site report written.

TrueSite: point a phone at a QR code and the BIM model appears over the real site at true scale. Tap **Check as-built**: Gemini compares what is built with the design, pins the deviations and drafts the issue.

Users: field engineers, foremen, site architects · Buyers: contractors

HOW IT WORKS

A link and a QR code. Nothing to install.



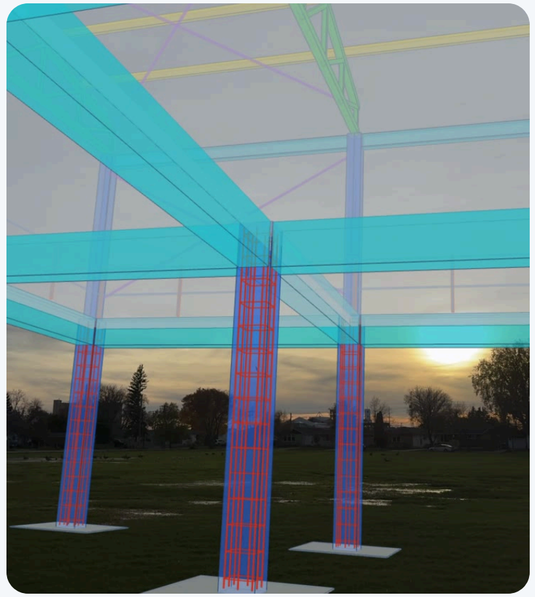
1 · Scan

The phone’s camera reads the code; TrueSite opens in the browser.



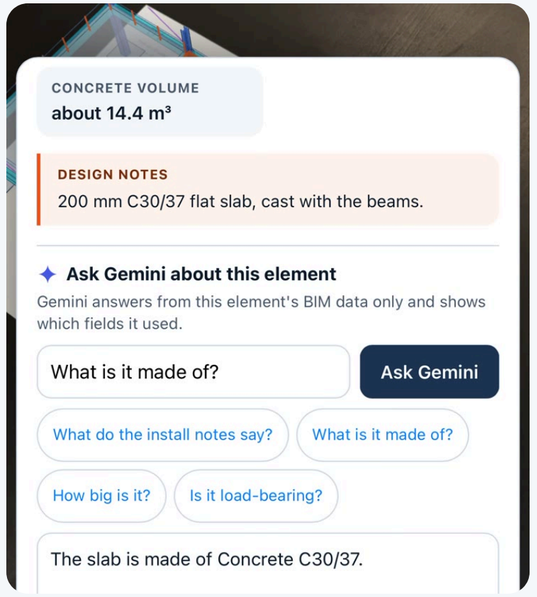
2 · Lock

Ten seconds of tilting calibrates the camera; the code’s corners give the pose.



3 · See

The design at 1:1, held by the motion sensors, levelled by gravity.



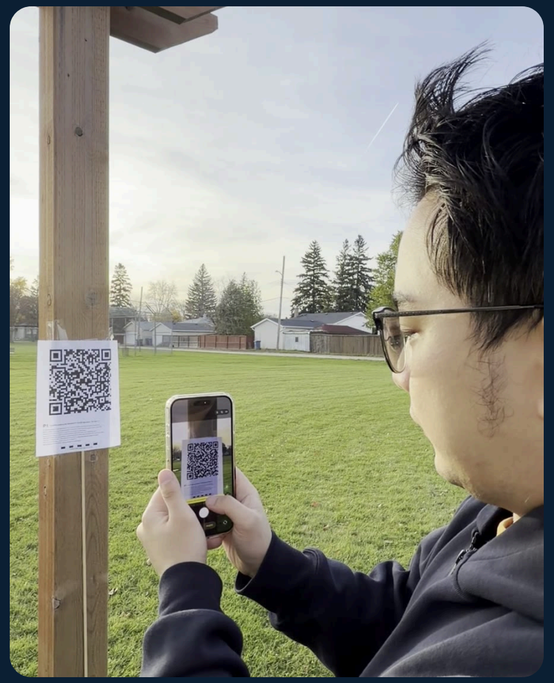
4 · Tap, ask, check

Any element’s BIM data; Gemini answers and checks the build.

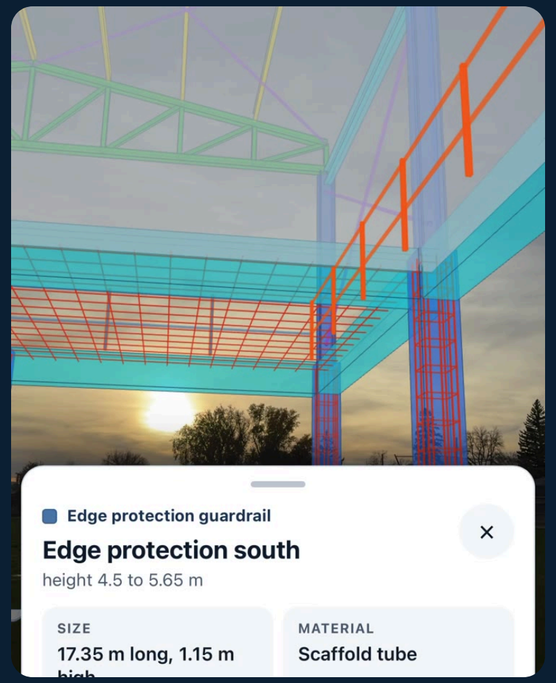
Screens from our iPhone field test, 17 Oct 2026, on the deployed Cloud Run service

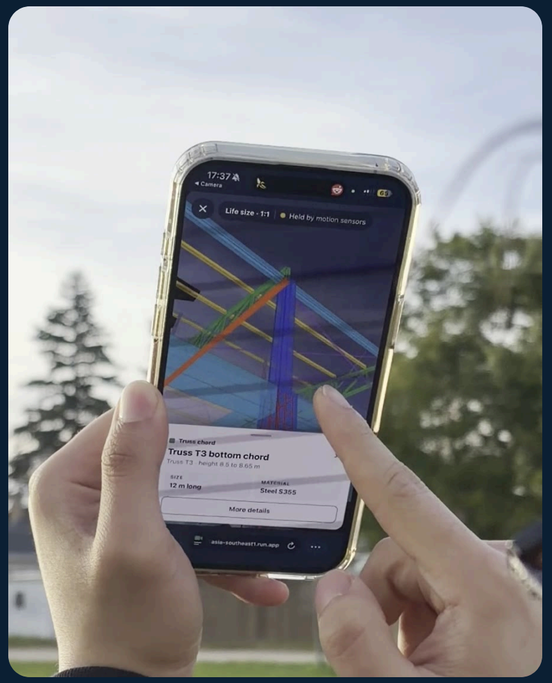
FIELD TEST · 17 OCT 2026

Life size on a real field, from an iPhone’s browser









The site sheets on posts put the building at 1:1; the desk sheet shows it at 1:100. The model held still while turning and looking up, and every element answered a tap with its BIM data.

Real footage on the deployed Cloud Run URL · no app, no headset · the iPad build fixed on 18 Oct, iPhone path unchanged

THE GEMINI AS-BUILT CHECK

Gemini judges. Code decides how serious.

1 · Three images

Your photo, the design rendered from the same pose with numbered marks, and the photo with each element’s outline

2 · Gemini 3.8 Flash

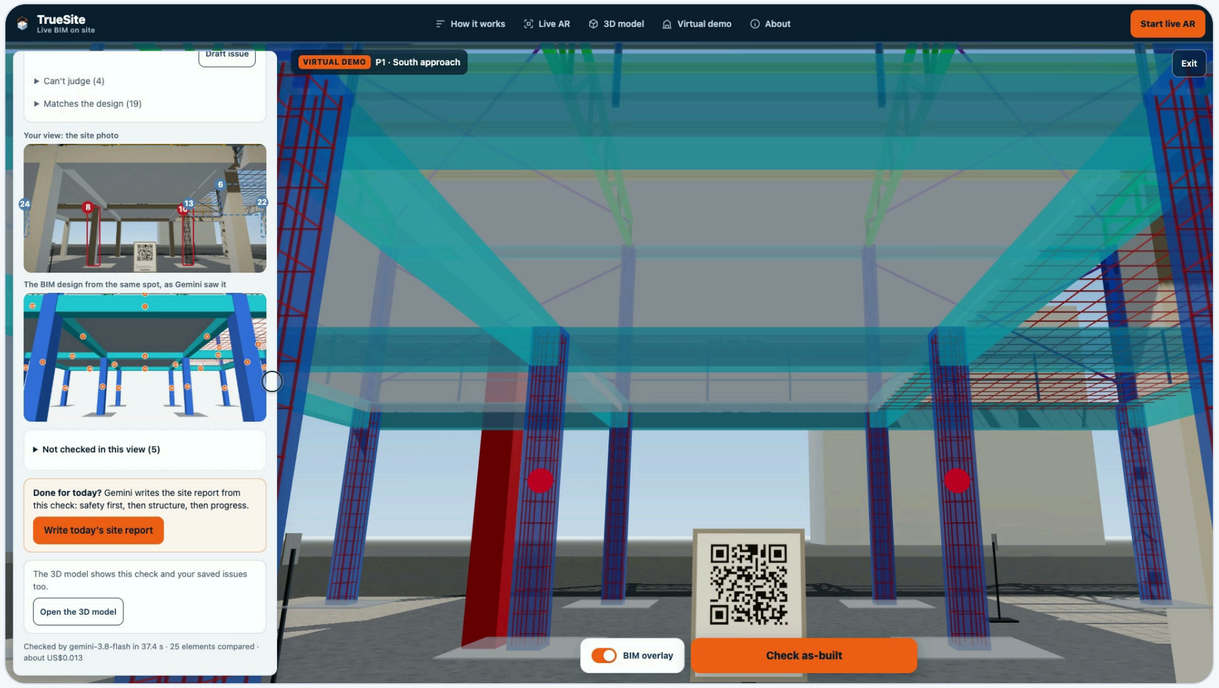
Judges each mark at its outline; returns structured JSON

3 · Server rules

One finding per mark, offsets in tens of cm, low confidence becomes “Can’t judge”; severity from the element class

4 · Pinned, explained, drafted

Pins on the model, “Why it matters” from fixed rules, the issue drafted by Gemini for a person to save

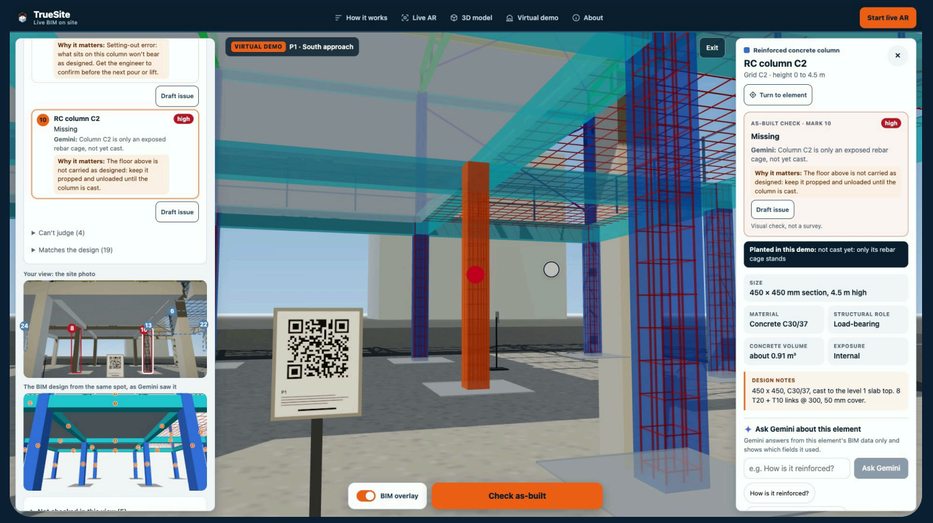


A live check in the virtual demo: the scorecard compares Gemini’s findings with the defects planted in view. Gemini never sees that list.

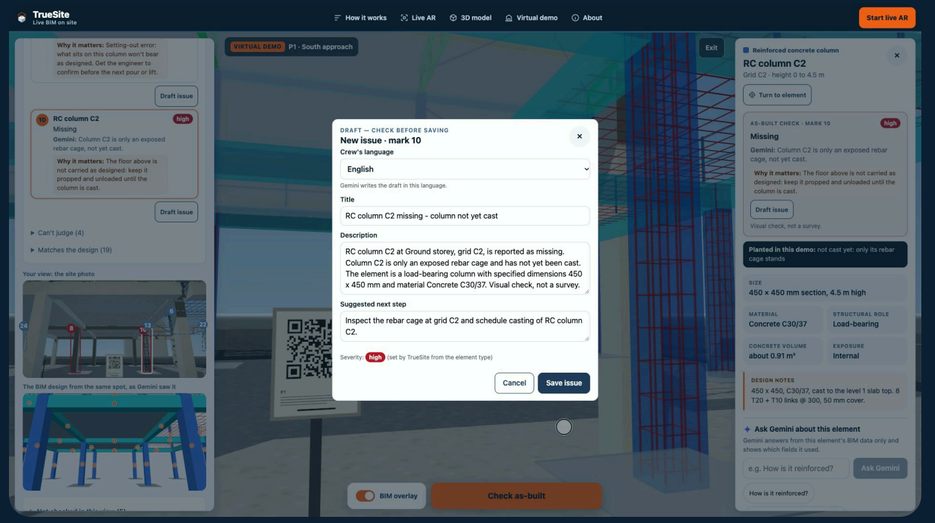
Every finding is labelled “Visual check, not a survey.” Gemini has no tools and no write access; its output is schema-checked.

ASK · DRAFT · SAVE

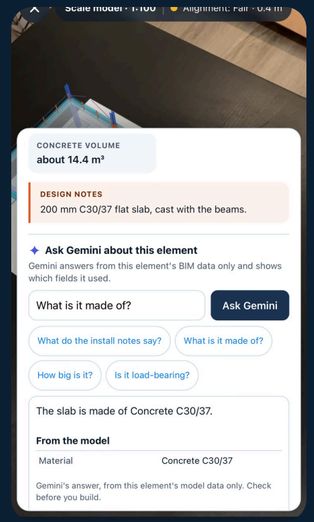
Gemini drafts. People decide.



Findings pinned on the view, each with “Why it matters” from fixed rules



Draft issue in one of eight crew languages; a person edits and saves it



Ask from the BIM data

Answers come only from the element’s own BIM properties, with the fields shown. Gemini also writes the day’s site report from the checks, safety first. Nothing Gemini writes is saved without a person.

Virtual demo: a simulated as-built site with 10 defects planted on purpose, so judges can try the check without a building

SOLUTION ARCHITECTURE · GOOGLE CLOUD

The phone does the vision. Google Cloud does the judging.

Phone browser

React + three.js, one web app

- QR reading in a zxing-wasm worker
- Pose from the code's corners; 10 s self-calibration
- Sensor fusion, gravity levelling
- Overlay at true scale
- ID buffer → numbered marks, design render, outlines

3 images

Cloud Run

asia-southeast1 · FastAPI and the web app on one origin

Limits and caps · schema repair · severity by rule · sessions, no accounts

Gemini 3.8 Flash

Agent Platform, global · structured JSON · busy or hung: 3.5 Flash, then Flash-Lite

Firestore

Check runs · issues · atomic daily cap

Cloud Storage

Photos · renders · 30-day lifecycle

Also: Cloud Build and Artifact Registry for source deploys · Secret Manager (API-key fallback only) · budget alerts · a least-privilege service account

A check costs about US$0.015 in Gemini tokens, measured on the deployed service

ROBUSTNESS · MEASURED, NOT CLAIMED

What we measured

Measurement	Result
Pose accuracy · simulation, 0.5 px corner noise, 0.5–3 m	median 3.6 mm, 0.66°
Self-calibration · simulation, 30 tilted views	focal length within 1%
Steadiness on a stand · ray-traced test videos	0.02–0.2 px jitter a frame (was 1–44 px)
As-built check · simulated site, 16 planted defects in view, 3 live runs	found 12, 11, 7 · false alarms 5, 8, 0
Cost of one check · deployed service	about US$0.015
Automated tests	549 web · 520 server · 88 BIM · 167 end-to-end

End to end on a laptop, an iPad and an iPhone profile, plus ray-traced camera videos with exact ground truth for live AR. Mutation, property and API fuzz testing on the core logic: 4,162 generated API cases, no server errors.

Simulation numbers are not field accuracy; the overlay error on a device at 1–3 m is not measured yet. We publish what we measure.

INNOVATION · PRIOR ART, OWNED

BIM on site exists. We add the check.

Already on the market

Dalux · Trimble · GAMMA AR · XYZ Reality

They place the model with markers or survey hardware and show it on site. Most need an installed app or dedicated hardware.

Gemini as-built check

Three aligned images, structured findings, severity by rule

Grounded Ask

Answers only from the element's own BIM data, fields shown

Drafted, not decided

Issues and site reports in the crew's language, saved by a person

A link and a QR code

No app, no headset; any phone; honest, measured accuracy

The new idea: render the design from the photo's exact pose, mark every visible element, and let Gemini judge each one against its own outline.

Built from scratch during the hackathon (repository started 11 Oct 2026, every commit dated)

BUSINESS CASE

Sold per field seat. Pennies to run.

Who pays

Main contractors and specialist subcontractors, per field seat, monthly. Proposed launch price: US$39 a seat a month, to validate in the pilot.

Cost to serve

About US$0.015 a check, measured. A heavy user at 20 checks a day for 22 days spends about US$6.60 of Gemini a month.

Singapore first

S$47–53 billion of construction demand projected for 2026, and BIM submission already mandatory over 5,000 m^2. Then Taiwan and the rest of JAPAC.

An 8-week pilot

One or two Singapore contractors, 10 field seats. Success: deviations caught before pouring, time to answer a model question on site, false-alarm rate.

Sources: Building and Construction Authority, Singapore (Jan 2026 outlook; BIM e-submission); check cost logged on our deployed service

IMPACT

Less rework. Faster sign-off. Safer sites.

Catch it before it’s poured

A missing cage or a shifted column found on site, while it can still be fixed, not weeks later as rework.

Safer for workers

Guardrail and edge-protection deviations are always rated high, and issues reach the crew in their own language.

Anyone on the crew

A phone and a printed code: no app, no headset. Subcontractors join by scanning.

Next

A real-site pilot with labelled photos to measure the check · world-locked AR on Android and walk-inside on iPhone (built behind a flag, not yet device-tested) · issue export to trackers · multi-code site set-up

Community benefit: safer sites and fewer wasted materials, for crews of every language

TEAM



Wai Lin

Product and code: live AR, pose and sensor fusion, the Gemini pipeline, Google Cloud



Aung Nyein Chan (Michael)

Architect: the BIM model, field validation and visuals

Try it: truesite-1067044308557.asia-southeast1.run.app

Watch: the demo video on the start page

Read the code: github.com/wl-geek/truesite

Thank you.

TrueSite

BIM ON SITE · AS-BUILT AI